

Practical - 10

Aim: Minimum Path Sum” says that given a $n \times m$ grid consisting of non-negative integers and we need to find a path from top-left to bottom right, which minimizes the sum of all numbers along the path.

PROGRAM CODE :

```
#include <stdio.h>
#include <limits.h>
#define MIN(a, b) ((a) < (b) ? (a) : (b))
int minPathSum(int grid[][100], int n, int m) {
    int dp[100][100];

    dp[0][0] = grid[0][0];

    for (int i = 1; i < m; i++) {
        dp[0][i] = dp[0][i - 1] + grid[0][i];
    }

    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + grid[i][0];
    }

    for (int i = 1; i < n; i++) {
        for (int j = 1; j < m; j++) {
            dp[i][j] = grid[i][j] + MIN(dp[i - 1][j], dp[i][j - 1]);
        }
    }

    return dp[n - 1][m - 1];
}
```

```
}  
int main() {  
    int grid[100][100];  
    int n, m;  
  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
    printf("Enter number of columns: ");  
    scanf("%d", &m);  
  
    printf("Enter the grid elements:\n");  
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < m; j++) {  
            scanf("%d", &grid[i][j]);  
        }  
    }  
  
    int result = minPathSum(grid, n, m);  
    printf("Minimum path sum: %d\n", result);  
    return 0;  
}
```

OUTPUT :-

Enter number of rows: 3

Enter number of columns: 3

Enter the grid elements:

1 3 1

1 5 1

4 2 1

Minimum path sum: 7

main.c	Output
<pre> 19 dp[i][j] = grid[i][j] + MIN(dp[i - 1][j], dp[i][j - 1]); 20 } 21 } 22 23 return dp[n - 1][m - 1]; 24 } 25 int main() { 26 int grid[100][100]; 27 int n, m; 28 29 printf("Enter number of rows: "); 30 scanf("%d", &n); 31 printf("Enter number of columns: "); 32 scanf("%d", &m); 33 34 printf("Enter the grid elements:\n"); 35 for (int i = 0; i < n; i++) { 36 for (int j = 0; j < m; j++) { 37 scanf("%d", &grid[i][j]); 38 } 39 } 40 41 int result = minPathSum(grid, n, m); </pre>	<pre> /tmp/ncpmRZbd1H.o Enter number of rows: 3 Enter number of columns: 3 Enter the grid elements: 1 3 1 1 5 1 4 2 1 Minimum path sum: 7 === Code Execution Successful === </pre>